

Highlights

Imaging-Informed Progressive Fusion for Mosaic-based Hyperspectral Image Reconstruction

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- An imaging-informed progressive reconstruction architecture is proposed for mosaiced and PAN image fusion.
- The complex mosaicing degradation is decomposed into hierarchical physically grounded sub-stages.
- Cross-scale parameter sharing and a three-stage composite loss are introduced to improve interpretability and reconstruction fidelity.
- Extensive experiments on simulated and real-world datasets demonstrate state-of-the-art performance.

Imaging-Informed Progressive Fusion for Mosaic-based Hyperspectral Image Reconstruction

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ABSTRACT

Reconstructing high-resolution (HR) hyperspectral images (HSIs) from low-resolution (LR) mosaiced and panchromatic (PAN) observations is a challenging, highly ill-posed inverse problem. Existing approaches often rely on black-box networks that lack physical interpretability, frequently leading to spatial artifacts and spectral distortion. In this paper, we propose an imaging-informed progressive fusion framework that explicitly models the hierarchical physics of mosaic degradation. Specifically, we decompose the complex reconstruction process into three physically grounded stages that sequentially invert the hierarchical sampling and pooling operations inherent in the mosaic imaging pipeline, transforming it into a series of tractable sub-tasks. By formulating a three-stage composite loss function, the proposed method effectively enforces both intermediate and observation consistency, preserving structural relationships across scales. Furthermore, to ensure coherent feature propagation, we introduce a cross-scale parameter-sharing strategy that exploits the scale-invariant nature of the sampling degradation. Extensive experiments on simulated and real-world datasets demonstrate that our proposed method achieves state-of-the-art performance.

1. Introduction

Hyperspectral imaging plays a transformative role in extending human perception by capturing continuous spectral signatures, which is fundamental to remote sensing [1], clinical diagnostics [2, 3], industrial inspection [4], etc. Conventional scanning-based modalities, however, are often plagued by sluggish acquisition speeds, making them ill-suited for monitoring dynamic targets. To address this temporal bottleneck, mosaic-based snapshot HSI has emerged as a compelling alternative. Drawing inspiration from the Bayer filter array used in RGB cameras, this technique employs a spectral filter array (SFA) to record spatial-spectral information within a single integration period. Such a mechanism facilitates video-rate HSI reconstruction [5, 6], providing a significant boost to temporal efficiency. Nevertheless, this advancement entails a fundamental trade-off, where spatial resolution is partially sacrificed to accommodate the diverse spectral bands.

As illustrated in Fig. 1(a), a feasible solution to mitigate this limitation is to fuse the mosaiced image with an auxiliary PAN image. Current research in this field is bifurcated into classical approaches [7, 8, 9, 10, 11, 12, 13] and deep learning (DL) architectures [14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24]. Traditional techniques generally hinge on handcrafted priors [7], such as sparse representation [10], matrix factorization [8, 12, 13], or tensor-based decomposition [9]. While these methods are characterized by mathematical rigor, their limited representation capacity

often fails to capture intricate spectral-spatial dependencies. Moreover, the heavy computational burden associated with these iterative solvers often restricts their real-time applicability.

In contrast, DL-based paradigms have set new performance benchmarks by learning direct mappings from large-scale datasets. These can be further categorized into supervised and unsupervised learning. Supervised frameworks [15, 16, 25, 26, 27, 28] necessitate HR-HSI ground truth for model optimization, which is notoriously difficult to acquire in practice. To address this data scarcity, unsupervised methods [17, 21, 22, 29, 30, 31] have emerged, bypassing the need for paired labels. A notable example is the equivariant fusion network (EFN) [21], which integrates the fusion process into a unified architecture. More recently, Dian *et al.* proposed an equivariant Bayesian variational inference framework (EBVIF) for mosaiced and PAN image fusion [22], demonstrating great performance under more complex condition. However, these existing unsupervised models are predominantly data-driven and overlook the explicit physical mechanisms of the imaging process. This lack of physical interpretability may compromise the fidelity of the reconstructed results.

Despite the promising results achieved by unsupervised and physics-guided deep learning approaches, a key limitation remains in their architectural design. Specifically, while many methods successfully incorporate the forward imaging model into loss functions to enforce data consistency [17, 21, 22, 30, 32], they typically rely on monolithic, one-step reconstruction networks. Some recent efforts have attempted to bridge this gap by embedding physical priors directly into the network topology [33, 34, 35] or by unfolding iterative optimization schemes into sequential operations [36, 37, 38, 39] to enable more structured mappings.

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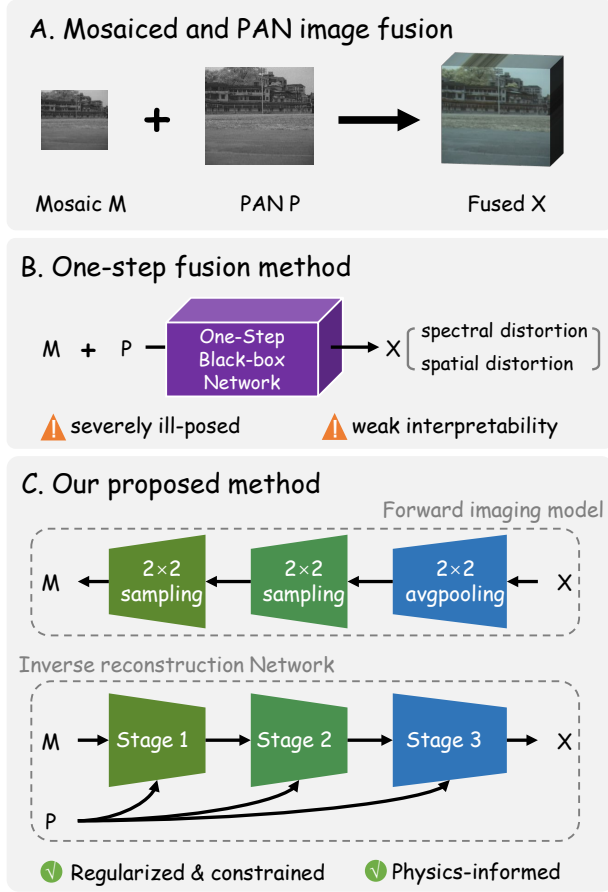


Figure 1: Illustration of the mosaiced and PAN image fusion task and representative one-step and the proposed imaging-informed fusion methods.

However, these architectures still predominantly treat the physical degradation as an atomic process. Consequently, they fail to explicitly account for the multi-stage nature of the physical imaging mechanism, leaving the hierarchical structure of the mosaicing degradation underexplored and the reconstruction process insufficiently interpretable.

Furthermore, the gap between the LR mosaiced input and the desired HR-HSI is too substantial to be effectively bridged in a single, monolithic step without introducing artifacts. Inspired by the multi-scale self-similarity inherent in natural scenes [40], we argue that the reconstruction should follow a hierarchical trajectory that mirrors the cascaded degradation process. Specifically, the relationship between different spatial scales in hyperspectral data often exhibits a structural recurrence, implying that the inverse mapping learned at a coarser scale can provide valuable inductive biases for finer scales [41]. Exploiting such cross-scale consistency not only reduces the optimization search space but also ensures that the reconstructed details across different resolutions remain coherent and interpretable.

Depicted in Fig. 1(c), we propose an imaging-informed progressive fusion framework for reconstructing HR HSI

from LR mosaiced observations and PAN images. To address the intrinsic ill-posedness of the fusion task, we explicitly decompose the forward imaging model into a hierarchical sequence of sampling and pooling operations. Mirroring this physical process, our network unfolds the reconstruction into three tractable stages, leveraging a parameter-sharing strategy to exploit cross-scale structural consistency. By integrating a three-stage composite loss function that incorporates intermediate and observation consistency, our framework ensures high-fidelity reconstruction and physical interpretability. The main contributions of this work are summarized as follows:

- We propose an imaging-informed progressive reconstruction architecture that bridges the gap between physical imaging mechanisms and deep learning. By reformulating the complex mosaicing degradation into a sequence of hierarchical operations, we transform the highly ill-posed inverse problem into a series of explicit and manageable sub-stages.
- We introduce a cross-scale parameter-sharing strategy and a three-stage composite loss function. By leveraging the structural recurrence inherent in hyperspectral data and enforcing consistency across intermediate representations, the proposed method enhances the interpretability of the fusion process while effectively regularizing the solution space.
- Extensive experiments on both simulated and real-world datasets demonstrate that the proposed framework achieves state-of-the-art performance.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the forward imaging model and the proposed imaging-informed progressive inverse reconstruction framework. Section 4 reports experimental results and ablation studies. Finally, Section 5 concludes the paper.

2. Related Work

2.1. Fusion Methods

As mentioned above, the mosaiced and PAN image fusion can be decomposed into the spectral demosaicing and pansharpening. Early spectral demosaicing methods [7, 8, 9, 10] predominantly relied on hand-crafted priors. A representative classical approach is the PPID algorithm [7], which achieves robust performance by leveraging a pseudo-panchromatic image estimated directly from the mosaiced input. In parallel, traditional pansharpening techniques exploited priors such as spectral-spatial consistency, sparse representation, and structural similarity [11, 12, 13]. However, these methods are often constrained by rigid hand-crafted priors, resulting in noticeable spectral or spatial distortions. Furthermore, most classical approaches rely on computationally intensive iterative optimization schemes, rendering them inefficient and impractical for real-world deployment.

Conversely, DL-based approaches deliver impressive fusion results owing to their powerful non-linear approximating capability, while maintaining high computational efficiency through streamlined single-pass inference. For instance, Liu *et al.* [19] proposed polarity memory network (PMNet) with quant attention to establish global correlation, thus reconstructing high-quality hyperspectral image from mosaiced image. Zhang *et al.* [16] designed a deep variational pansharpening network, namely VBPN, to yield superior results with enhanced interpretability and generalization. Huang *et al.* proposed wavelet-assisted fusion network (WFANet) for pansharpening [20], aiming to enable loss-less reconstruction. However, these supervised methods rely heavily on ground-truth HSI, which is usually inaccessible in the real-world scenario. Recent research has increasingly pivoted towards unsupervised paradigms. For example, Ma *et al.* [17] proposed a GAN-based framework (PanGAN) that enforces explicit cycle-consistency constraints to preserve both spectral and spatial fidelity. In parallel, Feng *et al.* [42] integrated an equivariant imaging prior into an unsupervised demosaicing architecture. More recently, Wang *et al.* proposed an unsupervised equivariant imaging (EI)-based fusion framework, namely EFN [21], for the mosaiced and PAN image fusion, demonstrating promising results. Furthermore, Dian *et al.* introduced the Bayesian variational inference framework named EBVIF [22], to make a more powerful fusion framework under the noisy scenarios.

2.2. Cross-scale Learning

Furthermore, cross-scale self-similarity, also referred to as structural recurrence in natural scenes, has been widely recognized as an effective internal prior for image reconstruction. For instance, Shocher *et al.* [40] introduced the zero-shot super-resolution framework, which exploits the recurrence of internal image statistics across scales to train an image-specific CNN at test time, thereby enabling super-resolution without relying on external training data. Building upon this idea, Zhou *et al.* [43] developed a cross-scale internal graph neural network that leverages the recurrence of similar patches across different resolutions to transfer high-resolution structural cues from downsampled images.

In the hyperspectral fusion literature, Dian *et al.* and Guo *et al.* [44, 45] first estimated the spatial blur kernel and then used the identified degradation process to construct self-supervised training pairs by progressively downsampling the observed LR HSI and HR multispectral image (MSI). Specifically, the original LR HSI, together with its further downsampled version and the correspondingly degraded HR MSI, were used to establish cross-scale training relationships. More recently, Guo *et al.* [41] further demonstrated that when the estimated blur kernel is recursively applied twice to the target HR HSI, producing two lower-resolution versions, two DNNs with identical architectures trained at these different scales achieve nearly equivalent performance when evaluated on the same reconstruction task. This finding suggests that reconstruction mappings across scales share strong structural consistency. Such an observation provides

an important motivation for our design, where cross-scale parameter sharing is introduced to explicitly exploit this intrinsic recurrence in hyperspectral reconstruction.

2.3. Physics-Guided Deep Reconstruction

Physics-guided deep reconstruction has recently attracted increasing attention in hyperspectral restoration and fusion, as it combines the representation ability of deep networks with the interpretability of physical imaging models. A common strategy is to incorporate the forward degradation process $\mathbf{y} = \mathcal{A}(\mathbf{x})$ into the training objective, where the reconstructed image is re-degraded and constrained to match the observed measurements. For example, Du *et al.* [30] introduced a self-supervised modality-decoupled fusion framework to disentangle shared and complementary information across modalities, while Chen *et al.* [32] coupled low-rank tensor priors with a self-supervised spatial-spectral guided network to exploit intrinsic spatial-spectral correlations without paired training data.

Some recent studies further improve structural interpretability by embedding physical or modality-aware priors into network architectures, or by unfolding optimization models into learnable stages. For instance, He *et al.* [34] exploits Tucker decomposition and spatial-spectral manifold learning for blind fusion, and Zhu *et al.* [35] combines Mamba with implicit neural representation for global perception and continuous-domain modeling. Meanwhile, unfolding-based methods [37, 39] transform physically motivated variational optimization into interpretable deep networks. Nevertheless, these methods are mainly driven by modality-adaptive fusion, tensor/manifold priors, continuous-domain representation, or generic optimization constraints, rather than by an explicit decomposition of the mosaicing-specific degradation mechanism.

3. Methodology

3.1. Overview

In this section, we present the proposed imaging-informed progressive fusion framework, designed to reconstruct a HR HSI from a LR mosaiced observation and a corresponding HR PAN image. To address the complex spatial-spectral coupling inherent in the mosaicing pattern, we first revisit the imaging model, reformulating the standard single-step degradation into a progressive sequence of average pooling and hierarchical sampling operations. Guided by this physical decomposition, we subsequently develop an imaging-informed progressive reconstruction network. By mirroring the cascaded degradation process, our network unfolds the highly ill-posed inverse problem into three explicit, tractable stages, leveraging a parameter-sharing strategy to ensure cross-scale consistency. Finally, to optimize this progressive architecture, we formulate a three-stage composite loss function. This objective not only constrains the intermediate representations for enhanced interpretability but also guarantees the high fidelity of the final fusion result by coupling observation consistency with an EI mechanism.

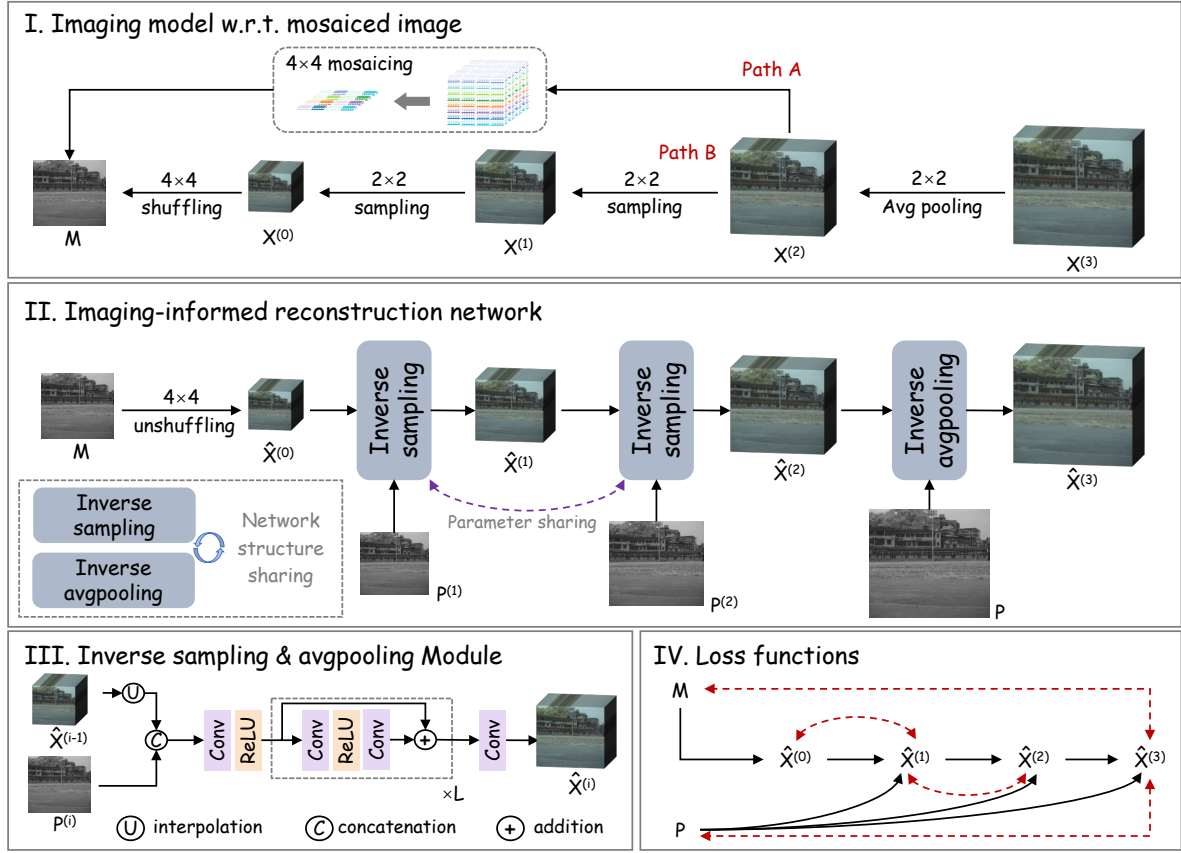


Figure 2: Illustration of the proposed fusion framework for the mosaiced and PAN image.

3.2. Imaging Model

Let $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ denote the target HR HSI, where H , W , and C represent the spatial height, width, and the number of spectral bands of the hyperspectral cube. The corresponding HR PAN image is denoted as $\mathbf{Y} \in \mathbb{R}^{1 \times H \times W}$, while the LR mosaiced image is denoted as $\mathbf{Z} \in \mathbb{R}^{1 \times h \times w}$, where $\frac{H}{h} = \frac{W}{w} = 8$.

The PAN image $\mathbf{Y}$ could be modeled as the spectral projection of the HR HSI:

$$\begin{aligned} \mathbf{Y} &= \mathcal{A}_Y(\mathbf{X}) + \mathbf{N}_Y \\ &= \mathbf{R}\mathbf{X} + \mathbf{N}_Y, \end{aligned} \quad (1)$$

where $\mathcal{A}_Y(\cdot)$ denotes the degradation operator w.r.t. the PAN image parameterized by the SRF, i.e., $\mathbf{R} \in \mathbb{R}^{1 \times C}$. $\mathbf{N}_Y$ denotes the composite noise and is omitted in the following formulation.

As illustrated in Fig. 2-I, acquisition of the mosaiced image $\mathbf{M}$ could be modeled as:

$$\begin{aligned} \mathbf{M} &= \mathcal{A}_M(\mathbf{X}) + \mathbf{N}_Z \\ &= D_{4 \times 4}(\mathbf{X})_{\downarrow 2 \times 2} + \mathbf{N}_Z, \end{aligned} \quad (2)$$

where $\mathcal{A}_M(\cdot)$ denotes the degradation operator for the mosaiced image. $D_{4 \times 4}(\cdot)$ represents the 4×4 mosaicing operation, and $(\cdot)_{\downarrow 2 \times 2}$ refers to the 2×2 average pooling. The additive noise $\mathbf{N}_Z$ can be ignored. We define this standard degradation process as Path A.

Existing fusion methods following Path A often overlook the inherent spatial-spectral coupling within the mosaicing pattern, attempting to solve the inverse problem in a single step. To address this, we propose a progressive fusion network that decomposes the mosaiced degradation into a sequence of sampling and pooling operations. Specifically, the 4×4 mosaicing is reformulated as a hierarchical sampling process, consisting of two successive 2×2 operations, i.e.,

$$\mathbf{X}^{(0)} = S_{2 \times 2}(S_{2 \times 2}((\mathbf{X})_{\downarrow 2 \times 2})), \quad (3)$$

where $\mathbf{X}^{(0)}$ denotes the downsampled hyperspectral cube and $S_{2 \times 2}(\cdot)$ represents the 2×2 sampling operation.

By applying a pixel-shuffle operation to $\mathbf{X}^{(0)}$, which rearranges 16 channels into a periodic 4×4 spatial pattern, we can derive the observed mosaiced image $\mathbf{M}$. Mathematically, this reformulated Path B is equivalent to Path A but provides a deeper structural decomposition, laying the foundation for our progressive fusion framework.

3.3. Imaging-Informed Progressive Reconstruction Network

Inspired by the decomposition of the forward imaging model, we construct an imaging-informed progressive reconstruction network, designed in a way that mirrors the degradation process. Illustrated in Fig. 2-II, this network consists of three stages, each implemented by a stage-wise

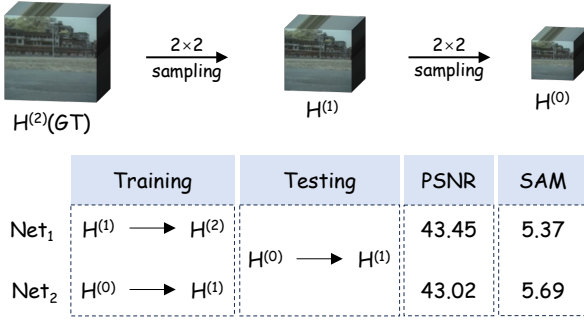


Figure 3: Illustration of the pilot study verifying the effectiveness of the cross-scale parameter-sharing strategy.

reconstruction module that learns to invert the corresponding forward degradation operation. The stage-wise reconstruction is expressed as:

$$\hat{\mathbf{X}}^{(i)} = \mathcal{H}_i(\hat{\mathbf{X}}^{(i-1)}, \mathbf{P}^{(i)}; \alpha_i), \quad i \in \{1, 2, 3\}, \quad (4)$$

where $\hat{\mathbf{X}}^{(i)}$ denotes the reconstructed HSI at i -th stage, and $\mathbf{P}^{(i)}$ represents the PAN image at the corresponding scale, downsampled via bilinear interpolation. $\mathcal{H}_i(\cdot; \alpha_i)$ represents the reconstruction module for the i -th stage, which is parameterized by α_i . $\hat{\mathbf{X}}^{(3)}$ is the expected fusion result. Specifically, $\mathcal{H}_1(\cdot)$ and $\mathcal{H}_2(\cdot)$ are designed to invert the 2×2 sampling operations, while $\mathcal{H}_3(\cdot)$ is dedicated to tackling the 2×2 average pooling degradation.

In general, the whole fusion process can be expressed as:

$$\hat{\mathbf{X}}^{(3)} = \mathcal{F}(\mathbf{M}, \mathbf{P}; \theta), \quad (5)$$

where $\mathcal{F}(\cdot; \theta)$ denotes the proposed imaging-informed progressive reconstruction network, parameterized by θ .

As depicted in Fig. 2-III, all reconstruction modules $\mathcal{H}_i$ share an identical architecture to ensure a consistent restoration process across scales. Taking the i -th stage as an example, the previous estimate $\hat{\mathbf{X}}^{(i-1)}$ is first upsampled by a factor of 2×2 to match the target spatial resolution. It is then concatenated with the scale-matched PAN image $\mathbf{P}^{(i)}$ along the spectral dimension, forming a $(C + 1)$ channel joint embedding that serves as the input to the module. The internal structure of $\mathcal{H}_i$ is designed for effective feature refinement. Specifically, an initial 3×3 convolutional layer maps the input to a 64-channel feature space, followed by a ReLU activation for non-linear spectral-spatial embedding. These features are then processed through three cascaded residual blocks, each comprising two 3×3 convolutions with 64 channels and an identity shortcut to facilitate gradient flow. Finally, a 3×3 convolutional layer projects the refined features back to C channels to produce the stage-wise reconstruction $\hat{\mathbf{X}}^{(i)}$. To maintain spatial resolution throughout the network, all convolutional layers employ a stride of 1×1 and padding of 1×1 .

As previously discussed, deep networks designed for certain image restoration tasks often exhibit cross-scale generalization, allowing them to be reused across different resolution levels. To verify the feasibility of a parameter-sharing strategy in our context, we conduct a pilot study. Specifically, we apply two sequential 2×2 sampling operations to an HR HSI, denoted as $\mathbf{H}^{(2)}$, yielding two downsampled versions, $\mathbf{H}^{(1)}$ and $\mathbf{H}^{(0)}$. We then construct two structurally identical 3-layer convolutional networks, denoted as Net₁ and Net₂. Net₁ is trained to reconstruct $\mathbf{H}^{(2)}$ from $\mathbf{H}^{(1)}$, whereas Net₂ is trained to reconstruct $\mathbf{H}^{(1)}$ from $\mathbf{H}^{(0)}$. During testing, both networks are evaluated on the $\mathbf{H}^{(1)} \rightarrow \mathbf{H}^{(2)}$ reconstruction task. As illustrated in Fig. 3, Net₂ achieves nearly identical reconstruction performance to Net₁ despite being trained at a different scale. This confirms the scale-invariant nature of the sampling degradation. Consequently, we adopt a parameter-sharing strategy for $\mathcal{H}_1(\cdot)$ and $\mathcal{H}_2(\cdot)$. Conversely, while $\mathcal{H}_3(\cdot)$ shares the same architecture as the first two modules, its weights are learned independently, as it is dedicated to inverting average pooling rather than spatial subsampling.

3.4. Loss Functions

As illustrated in Fig. 2-IV, we adopt a three-stage composite loss function as the overall loss function:

$$\mathcal{L} = \alpha_1 \mathcal{L}_{stage}^{(1)} + \alpha_2 \mathcal{L}_{stage}^{(2)} + \alpha_3 \mathcal{L}_{stage}^{(3)}, \quad (6)$$

where α_1 , α_2 , and α_3 are the weighting coefficients associated with the objectives of each stage. The terms $\mathcal{L}_{stage}^{(1)}$ and $\mathcal{L}_{stage}^{(2)}$ denote the inter-stage consistency losses for the first two reconstruction stages, while $\mathcal{L}_{stage}^{(3)}$ corresponds to the final-stage objective.

To construct $\mathcal{L}_{stage}^{(1)}$, the output $\hat{\mathbf{X}}^{(1)}$ of the first stage is constrained to be consistent with the initial estimate $\hat{\mathbf{X}}^{(0)}$, which preserves the original measurement information, after applying the subsampling operator corresponding to the second stage $\mathcal{S}_{2 \times 2}^{(2)}$, i.e., the second sampling operator in the cascaded mosaicing process:

$$\mathcal{L}_{stage}^{(1)} = \left\| \mathcal{S}_{2 \times 2}^{(2)}(\hat{\mathbf{X}}^{(1)}) - \hat{\mathbf{X}}^{(0)} \right\|_2^2. \quad (7)$$

Similarly, $\mathcal{L}_{stage}^{(2)}$ is imposed between the output $\hat{\mathbf{X}}^{(2)}$ and its corresponding input $\hat{\mathbf{X}}^{(1)}$, by enforcing consistency under the first sampling operator $\mathcal{S}_{2 \times 2}^{(1)}$, which corresponds to the inner sampling step in the cascaded degradation process:

$$\mathcal{L}_{stage}^{(2)} = \left\| \mathcal{S}_{2 \times 2}^{(1)}(\hat{\mathbf{X}}^{(2)}) - \hat{\mathbf{X}}^{(1)} \right\|_2^2. \quad (8)$$

To construct $\mathcal{L}_{stage}^{(3)}$, we follow [21, 22] and adopt the loss function as:

$$\mathcal{L}_{stage}^{(3)} = \lambda_m \mathcal{L}_{mosaic} + \lambda_p \mathcal{L}_{pan} + \lambda_{ei} \mathcal{L}_{ei}, \quad (9)$$

where λ_m , λ_p , and λ_{ei} are the corresponding weighting coefficients. Specifically, $\mathcal{L}_{mosaic}$ and $\mathcal{L}_{pan}$ denote the data

fidelity term, measuring the consistency with the observations, i.e., mosaiced and PAN images. $\mathcal{L}_{ei}$ represents the EI-based term, which can measure the distortion induced by the incapability of the fusion network.

The $\mathcal{L}_{mosaic}$ and $\mathcal{L}_{pan}$ are constructed respectively as:

$$\mathcal{L}_{pan} = \left\| \mathbf{P} - \mathcal{A}_P(\hat{\mathbf{X}}^{(3)}) \right\|_2^2, \quad (10)$$

$$\mathcal{L}_{mosaic} = \left\| \mathbf{M} - \mathcal{D}_{4 \times 4} \left((\hat{\mathbf{X}}^{(3)})_{12 \times 2} \right) \right\|_2^2. \quad (11)$$

Furthermore, the $\mathcal{L}_{ei}$ could be derived as:

$$\mathcal{L}_{ei} = \left\| \hat{\mathbf{X}}^{(3)} - \mathcal{F} \left(\mathcal{A}_M(\mathcal{T}_g(\hat{\mathbf{X}}^{(3)})), \mathcal{A}_P(\mathcal{T}_g(\hat{\mathbf{X}}^{(3)})); \theta \right) \right\|_2^2, \quad (12)$$

where $\mathcal{T}_g(\cdot)$ denotes the combination of spatial transformations, such as rotation, flipping, and translation.

Above all, we here detail the specific function of each loss term. The terms $\mathcal{L}_{stage}^{(1)}$ and $\mathcal{L}_{stage}^{(2)}$ constrain the intermediate representations during the fusion process, which enhances interpretability and thereby boosts the final reconstruction performance. Furthermore, $\mathcal{L}_{stage}^{(3)}$ ensures the high fidelity of the fusion result. Specifically, $\mathcal{L}_{mosaic}$ and $\mathcal{L}_{pan}$ enforce observation consistency, whereas $\mathcal{L}_{ei}$ enables the fusion network to effectively learn beyond the range space.

4. Experiments

4.1. Datasets

We evaluate the proposed method on two simulated datasets, i.e., CAVE [46] and Chikusei [47], together with a real-world dataset [21].

CAVE dataset contains 32 hyperspectral scenes, each with a spatial size of 512×512 and 31 spectral channels. In our setup, only a subset of contiguous bands, from the 12th to the 27th, is retained to form a 16-channel HR HSI. To simulate the corresponding observations, the hyperspectral data are first spatially reduced using 2×2 averaging, and then rearranged through an 4×4 mosaicing operation. The PAN image is generated via spectral projection using a predefined SRF, implemented as a weighted summation across spectral bands. Specifically, the SRF is defined as $[1, 1, 2, 4, 8, 9, 10, 12, 16, 12, 10, 9, 7, 3, 2, 1] / 107$, which is normalized to sum to one. For experimental evaluation, the dataset is partitioned into 20 samples for training and 12 samples for testing.

Chikusei dataset is captured over Chikusei, Japan, employing the Hyperspec-VNIR-CIRIS airborne sensor. This dataset consists of 128 spectral channels with a spatial resolution of 2.5 m and a grid size of 2335×2517 . Consistent with the band selection strategy used for the CAVE dataset, a 16-band HSI was extracted for our experiments. For numerical stability, pixel intensities were normalized by a factor of 2^{14} . Regarding the data partition, the original scene was divided along the width into a training sub-cube

of $2335 \times 2117 \times 16$ and a testing strip of $2335 \times 400 \times 16$. The latter was further segmented along the height to generate five 400×400 and one 335×400 test samples.

Real-world dataset is composed of 60 aligned image pairs, each including a mosaiced image and the corresponding PAN image, both captured using a custom-built dual-sensor imaging system. The two sensors are carefully calibrated to observe the same scene from an identical view-point, ensuring pixel-wise alignment. All acquisitions are performed under ambient daylight conditions, and no additional preprocessing is required for alignment. Among the collected samples, 50 pairs are used for training, while the remaining 10 pairs are reserved for evaluation.

4.2. Training Setup

The experimental setup and optimization strategy for our model are summarized below. Specifically, PAN patches are cropped with a spatial size of 64×64 , while the corresponding mosaiced patches are extracted according to the spatial resolution ratio, resulting in a size of 32×32 . The model parameters, including both the fusion network and the PAN degradation module, are optimized using the Adam optimizer with an initial learning rate of 10^{-4} and default momentum coefficients $(\beta_1, \beta_2) = (0.9, 0.999)$. We adopt a mini-batch size of 16 for all datasets. The training schedule varies across datasets to reflect their scale: the number of iterations is set to 2.8×10^5 for CAVE, 1.5×10^4 for Chikusei, and 3.3×10^4 for the real-world dataset. The entire framework is implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU.

4.3. Competing Methods and Evaluation Metrics

Due to the scarcity of existing studies that directly address mosaiced and PAN image fusion, we construct a set of two-stage baselines by decomposing the problem into spectral demosaicing followed by the LR HSI and PAN image fusion. Following this idea, a series of comparisons is established by pairing representative demosaicing algorithms with established fusion networks, covering both traditional and deep learning-based approaches. In particular, PPID [7], LSAN [42], and PMNet [19] are adopted for spectral reconstruction, while VBPN [16], PanGAN [17], and WFANet [20] are employed as fusion models, leading to multiple baseline combinations. These baselines follow a two-stage pipeline, where demosaicing and fusion are performed sequentially. In addition, EFN [21] and EBVIF [22] are included as one-stage approaches that directly perform joint reconstruction from mosaiced and PAN observations. For a consistent comparison, all methods are trained using the same configuration, including optimizer settings, learning rate, batch size, and patch sampling strategy.

Quantitative evaluation on simulated datasets is performed using full-reference metrics, where ground-truth HR-HSI is available. For consistency, PSNR, SSIM, SAM, and ERGAS [48] are employed across both the CAVE and Chikusei datasets. For real-world experiments, where reference data are not accessible, we instead utilize no-reference metrics, including QNR [49], D_λ , and D_S , to

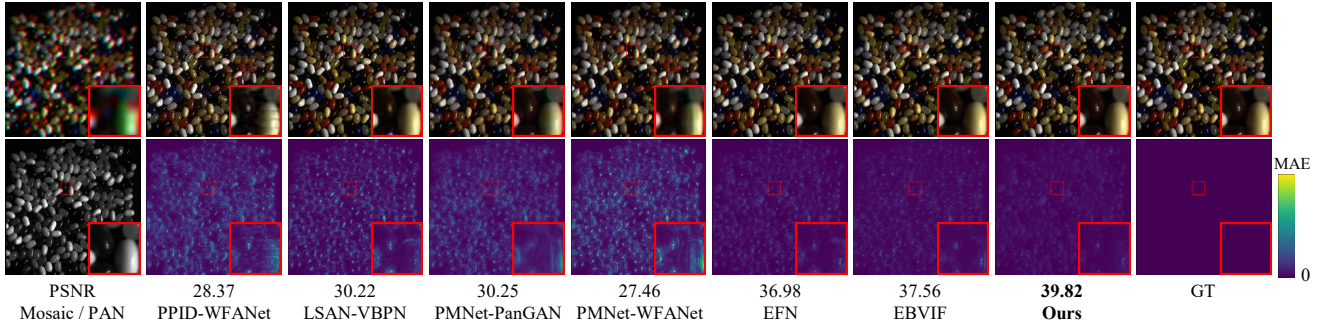


Figure 4: Qualitative comparison on the CAVE dataset. From left to right are representative competing methods and the proposed method (Ours). Red boxes highlight local regions for comparison.

assess overall fusion performance as well as spectral and spatial distortions. To compute D_λ in this context, the mosaiced measurements are first reorganized into an LR-HSI format to align with the original metric formulation.

4.4. Comparison Experiments

(1) Qualitative evaluation on simulated datasets. On the CAVE dataset, PPID-WFANet exhibits evident spatial artifacts, indicating its limited ability to recover fine spatial structures from degraded observations. PMNet-PanGAN and PMNet-WFANet generate visually sharper results, but they still introduce slight spectral discrepancies, which can be observed from the color inconsistency with the ground truth. LSAN-VBPN alleviates part of the spectral distortion but tends to produce mild spatial blurring, leading to the loss of local details. In contrast, EFN, EBVIF, and the proposed method achieve more balanced reconstruction results in terms of both spatial detail preservation and spectral fidelity. Notably, the proposed method produces clearer spatial textures and more faithful spectral responses, and obtains lower MAE and higher PSNR than the competing methods, demonstrating its superior reconstruction capability.

On the Chikusei dataset, the performance gap among different methods becomes more obvious. PPID-WFANet, LSAN-VBPN, PMNet-PanGAN, and PMNet-WFANet all suffer from severe spectral distortions, resulting in apparent color discrepancies compared with the ground truth. These distortions suggest that directly combining demosaicing and pansharpening modules may lead to accumulated reconstruction errors, especially under more complex spatial-spectral distributions. By comparison, EFN, EBVIF, and the proposed method provide more faithful visual results with reduced color bias and better structural consistency. Furthermore, the proposed method achieves lower MAE and higher PSNR than EFN, yielding the best overall qualitative and quantitative performance on the Chikusei dataset.

(2) Quantitative evaluation on simulated datasets. The quantitative results are reported in Table 1 and Table 2. On the CAVE dataset, the proposed method achieves the best performance across all evaluation metrics, obtaining 41.84 dB in PSNR, 0.9880 in SSIM, 4.6789 in SAM, and 1.9116 in ERGAS. Compared with the strongest competing method, EBVIF, our method improves PSNR by 0.92 dB and

reduces ERGAS by 0.3633. In comparison with EFN, the proposed method also brings a 1.29 dB PSNR improvement and achieves better SSIM, SAM, and ERGAS values. These results demonstrate that the proposed method can more effectively preserve spatial structures while maintaining high spectral fidelity.

On the Chikusei dataset, the proposed method also delivers the best overall quantitative performance. Specifically, it achieves the highest PSNR of 40.53 dB, the highest SSIM of 0.9788, and the lowest ERGAS of 3.9215 among all compared methods. Compared with EFN, our method improves PSNR by 0.10 dB, SSIM by 0.0009, and ERGAS by 0.1077, indicating slightly better spatial reconstruction accuracy and global fidelity. Although EFN obtains the best SAM value, the proposed method achieves a very competitive SAM of 16.1213, only slightly higher than EFN. Overall, these results suggest that our method provides a better balance between spatial detail preservation, structural consistency, and spectral reconstruction on the more challenging Chikusei dataset.

(3) Qualitative evaluation on real-world dataset. Fig. 6 presents the visual comparison results on the real-world dataset. It can be observed that PPID-WFANet, PMNet-PanGAN, and PMNet-WFANet suffer from obvious reconstruction artifacts, which severely degrade the visual quality of the recovered images. LSAN-VBPN alleviates part of these artifacts but introduces evident spatial blurring, leading to the loss of fine structural details. Although EFN and EBVIF achieve better spatial consistency than the above methods, they still suffer from noticeable spectral distortions. Specifically, EFN produces an undesirable color shift, while EBVIF tends to generate over-smoothed results with a grayish color appearance, indicating insufficient spectral fidelity in real-world scenarios. In contrast, the proposed method reconstructs more visually faithful results, preserving clearer spatial structures and more natural spectral responses. This demonstrates the stronger generalization ability of our method for real-world mosaiced and PAN image fusion.

(4) Quantitative evaluation on real-world dataset. The quantitative results are reported in Table 3. The proposed

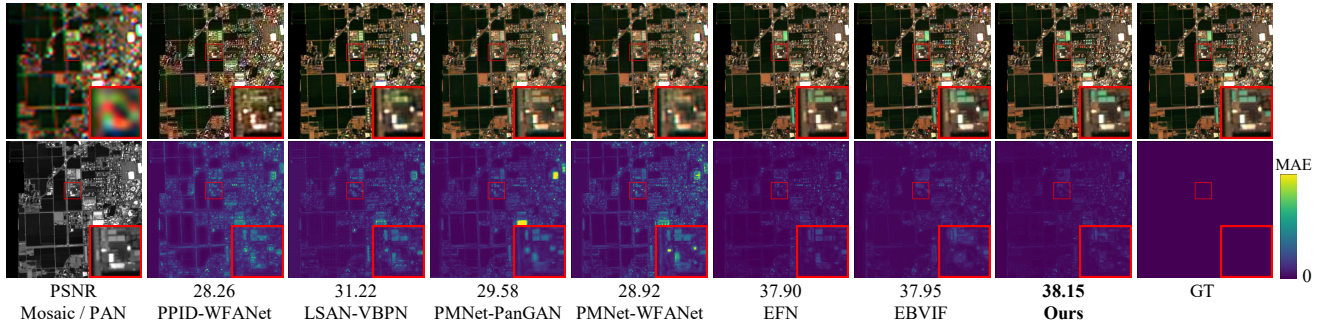


Figure 5: Qualitative comparison on the Chikusei dataset. From left to right are representative competing methods and the proposed method (Ours). Red boxes highlight local regions for comparison.

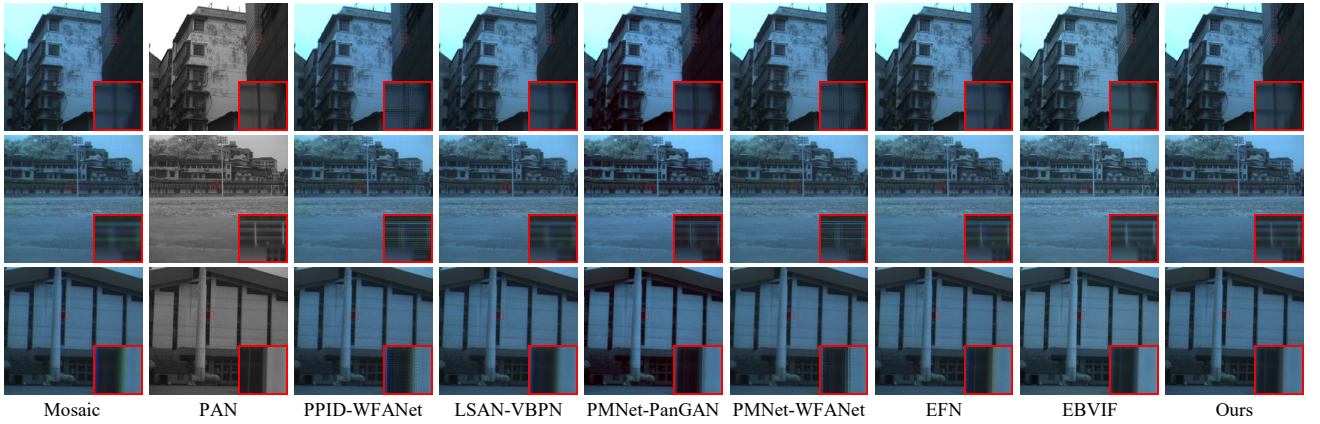


Figure 6: Qualitative comparison on the real-world dataset. From left to right are representative competing methods and the proposed method (Ours). Red boxes highlight local regions for comparison.

method achieves the highest QNR of 0.9186 and the lowest spatial distortion D_S of 0.0175 among all compared methods, demonstrating its superior overall fusion quality and strong capability in preserving spatial details. Compared with EFN and EBVIF, our method improves QNR from 0.8764 and 0.8902 to 0.9186, respectively, while further reducing D_S from 0.0226 and 0.0241 to 0.0175. These results indicate that the proposed method can more effectively enhance spatial fidelity and suppress structural distortion in real-world reconstruction.

4.5. Ablation Study on Loss Function Design

In addition to the proposed loss function configuration, we investigate four alternative designs, as illustrated in Fig. 7. Specifically:

- Configuration (a): The baseline setup that computes the loss solely between the final reconstruction $\hat{\mathbf{X}}^{(3)}$ and the two observations, i.e., $\mathbf{M}$ and $\mathbf{P}$, along with the EI-based loss function.
- Configuration (b): Extends (a) by anchoring all intermediate reconstructions to the final estimate $\hat{\mathbf{X}}^{(3)}$, by enforcing consistency between $\hat{\mathbf{X}}^{(1)}$ and $\hat{\mathbf{X}}^{(3)}$, and between $\hat{\mathbf{X}}^{(2)}$ and $\hat{\mathbf{X}}^{(3)}$.

- Configuration (c): Extends (a) by anchoring all intermediate reconstructions to the initial estimate $\hat{\mathbf{X}}^{(0)}$, enforcing consistency between $\hat{\mathbf{X}}^{(0)}$ and $\hat{\mathbf{X}}^{(1)}$, and between $\hat{\mathbf{X}}^{(0)}$ and $\hat{\mathbf{X}}^{(2)}$.
- Configuration (d): Extends (a) by introducing pairwise consistency losses between consecutive stages, i.e., between $\hat{\mathbf{X}}^{(0)}$ and $\hat{\mathbf{X}}^{(1)}$, between $\hat{\mathbf{X}}^{(1)}$ and $\hat{\mathbf{X}}^{(2)}$, and between $\hat{\mathbf{X}}^{(2)}$ and $\hat{\mathbf{X}}^{(3)}$.

Quantitative results on the real-world dataset are summarized in Table 4. Compared to configuration (a), our proposed design achieves consistently better performance, demonstrating the effectiveness of incorporating intermediate-stage supervision guided by the physical imaging process. When compared to configurations (b) and (c), our method also shows clear advantages. This suggests that directly anchoring intermediate outputs to either the initial estimate $\hat{\mathbf{X}}^{(0)}$ or the final target $\hat{\mathbf{X}}^{(3)}$, without respecting the hierarchical degradation structure, introduces error propagation and degrades overall reconstruction fidelity. Notably, even against configuration (d), which appears structurally similar but adds an extra consistency term between $\hat{\mathbf{X}}^{(2)}$ and $\hat{\mathbf{X}}^{(3)}$, our proposed loss formulation still yields superior results. We attribute this to the fact that the transition from $\hat{\mathbf{X}}^{(2)}$ to $\hat{\mathbf{X}}^{(3)}$ involves inverting an average pooling operation,

Table 1

Quantitative evaluation of the competing methods on the CAVE dataset. $\uparrow$ stands for the greater value the better, while $\downarrow$ stands for the smaller value the better. The best result of each quality index is marked in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$
PPID-VBPN	33.21	0.9007	10.7391	4.7555
PPID-PanGAN	34.85	0.9457	10.8829	4.0853
PPID-WFANet	31.59	0.8829	9.9394	5.6134
LSAN-VBPN	35.19	0.9334	9.0635	3.7457
LSAN-PanGAN	37.10	0.9684	7.7853	3.1186
LSAN-WFANet	35.30	0.9497	7.6442	3.6943
PMNet-VBPN	31.89	0.8878	10.3299	5.3833
PMNet-PanGAN	33.66	0.9370	11.4522	4.5036
PMNet-WFANet	31.70	0.8936	10.8467	5.5128
EFN	40.55	0.9833	5.3147	2.2556
EBVIF	40.92	0.9762	7.2797	2.2749
Ours	41.84	0.9880	4.6789	1.9116

Table 2

Quantitative evaluation of the competing methods on the Chikusei dataset. $\uparrow$ stands for the greater value the better, while $\downarrow$ stands for the smaller value the better. The best result of each quality index is marked in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	SAM $\downarrow$	ERGAS $\downarrow$
PPID-VBPN	33.37	0.8800	21.8146	8.1733
PPID-PanGAN	33.25	0.8990	23.2002	8.3579
PPID-WFANet	30.64	0.8124	24.3902	11.1093
LSAN-VBPN	33.67	0.8476	19.4292	7.7414
LSAN-PanGAN	33.77	0.9284	20.8048	7.7552
LSAN-WFANet	32.65	0.8938	20.8070	8.6556
PMNet-VBPN	32.03	0.8480	19.0318	9.3009
PMNet-PanGAN	33.12	0.8946	21.0539	8.3405
PMNet-WFANet	31.58	0.8366	20.7985	9.7755
EFN	40.43	0.9779	15.9419	4.0292
EBVIF	39.81	0.9701	16.9215	4.2100
Ours	40.53	0.9788	16.1213	3.9215

which is fundamentally different from the spatial subsampling operations addressed in earlier stages. Imposing a direct reconstruction loss between these two stages, as in configuration (d), fails to account for this distinction and may misguide the learning process. In contrast, our design could ensure both physical plausibility and robust generalization.

4.6. Ablation Study on Cross-Scale PAN Loss

We further investigate the effectiveness of incorporating a cross-scale PAN loss into our framework. Specifically,

Table 3

Quantitative evaluation of the competing methods on the real-world dataset. $\uparrow$ stands for the greater value the better, while $\downarrow$ stands for the smaller value the better. The best result of each quality index is marked in **bold**.

Method	QNR $\uparrow$	D_λ $\downarrow$	D_S $\downarrow$
PPID-VBPN	0.7664	0.0271	0.0765
PPID-PanGAN	0.8497	0.0323	0.0424
PPID-WFANet	0.7128	0.0439	0.0935
LSAN-VBPN	0.7557	0.0351	0.0785
LSAN-PanGAN	0.7620	0.0893	0.0579
LSAN-WFANet	0.6574	0.1283	0.0905
PMNet-VBPN	0.7556	0.0148	0.0848
PMNet-PanGAN	0.7981	0.0631	0.0523
PMNet-WFANet	0.7141	0.0400	0.0944
EFN	0.8764	0.0614	0.0226
EBVIF	0.8902	0.0420	0.0241
Ours	0.9186	0.0313	0.0175

Table 4

Quantitative assessment of different loss function configurations on the real-world dataset. The symbols $\uparrow$ and $\downarrow$ indicate that higher and lower values correspond to better performance, respectively. The best result for each metric is highlighted in **bold**.

Configuration	QNR $\uparrow$	D_λ $\downarrow$	D_S $\downarrow$
(a)	0.9147	0.0248	0.0211
(b)	0.8935	0.0364	0.0248
(c)	0.9147	0.0309	0.0190
(d)	0.9095	0.0240	0.0232
Ours	0.9186	0.0313	0.0175

we downsample the final reconstructed HSI $\hat{\mathbf{X}}^{(3)}$ to multiple intermediate spatial scales and apply the learned SRF, denoted as $\hat{\mathbf{R}}$, to generate corresponding PAN estimates at each scale. We then compute the reconstruction loss between these synthesized PAN images and their ground-truth counterparts obtained by downsampling the original HR PAN image $\mathbf{P}$. As shown in Fig. 8, adding this cross-scale PAN loss consistently degrades performance across most metrics on both the CAVE and real-world datasets, except for D_S . We attribute this degradation to the inaccurate supervision signal induced by bilinearly downsampled PAN images. That is, such approximations distort high-frequency spatial structures and misrepresent the actual relationship between scales, thereby imposing misleading constraints during training. The experimental results confirm that introducing auxiliary losses based on interpolated multi-scale PAN references is counterproductive, reinforcing our design choice to exclude such terms.

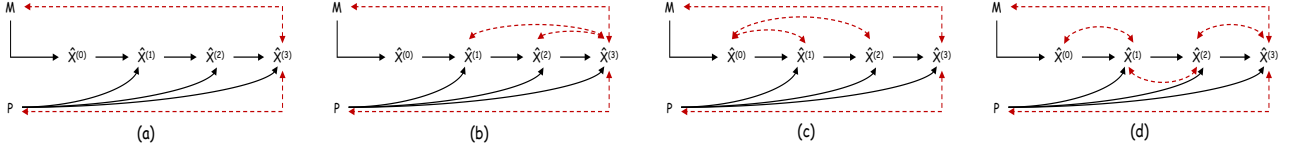


Figure 7: Schematic illustration of the four loss function configurations evaluated in the ablation study.

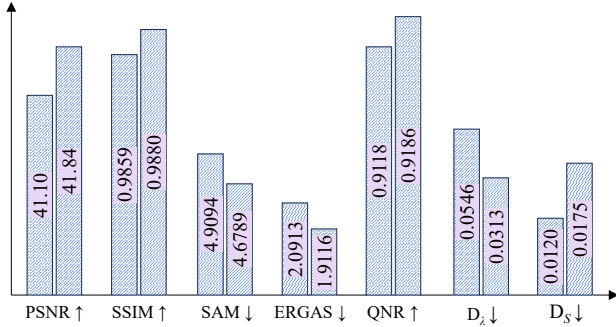


Figure 8: Quantitative evaluation of the impact of the cross-scale PAN-based constraint on the CAVE and real-world datasets. Note that the metrics PSNR, SSIM, SAM, and ERGAS are evaluated on the CAVE dataset, whereas QNR, D_λ , and D_S are measured on the real-world dataset.

4.7. Effect of the Cross-Scale Parameter Sharing

To validate effectiveness of the cross-scale parameter-sharing strategy adopted in our progressive reconstruction network, we conduct ablation experiments on the real-world dataset. Specifically, we compare our full model, which shares parameters between $\mathcal{H}_1(\cdot)$ and $\mathcal{H}_2(\cdot)$ to ensure consistent restoration across scales, against a variant where each stage-wise module is independently parameterized.

Qualitative results are presented in Fig. 9. As shown, the variant without parameter sharing exhibits noticeable spectral distortions and spatial artifacts in the fusion results. Without parameter sharing, each reconstruction module learns scale-specific mappings that may not generalize well across different resolution levels, leading to misalignment in the progressive refinement process. In contrast, our parameter-sharing design enforces a unified restoration behavior across scales, effectively leveraging the scale-invariant nature of the sampling degradation, as verified in our pilot study, and ensuring coherent feature propagation throughout the three-stage reconstruction pipeline. This ablation confirms that cross-scale parameter sharing is not merely a parameter-efficient design choice, but a critical component for achieving high-fidelity fusion.

4.8. Validation of Hierarchical Sampling Hypothesis

We verify the effect of our core hypothesis that decomposes the 4×4 mosaicing process into a sequence of two consecutive 2×2 sampling operations. A natural question arises: why not directly model the 4×4 mosaicing as a single-step operation? To address this concern and validate

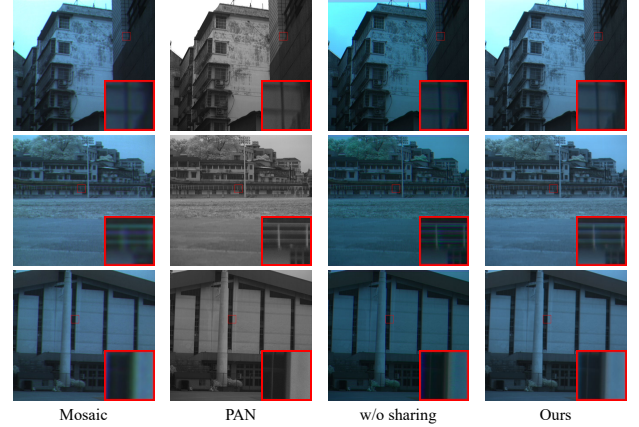


Figure 9: Visual comparison demonstrating the effect of omitting versus utilizing the proposed cross-scale parameter-sharing strategy on the real-world dataset.

our hierarchical decomposition strategy, we conduct comparative experiments on the real-world dataset. Specifically, we train an alternative model that treats the 4×4 mosaicing as an atomic, non-decomposable operation, where the network attempts to invert the entire 4×4 sampling in a single stage without intermediate supervision.

As shown in Table 5, the direct 4×4 approach achieves significantly inferior performance compared to our proposed hierarchical method. The single-step 4×4 model obtains a QNR of only 0.7856, substantially lower than our method's 0.9186, indicating severe degradation in overall fusion quality. More critically, both spectral distortion, $D_\lambda = 0.1212$, and spatial distortion, $D_S = 0.0374$, are dramatically worse than our approach, where $D_\lambda = 0.0313$ and $D_S = 0.0175$. This performance gap demonstrates that attempting to bridge the substantial resolution gap between the LR mosaiced input and HR HSI target in a single step introduces significant artifacts and reconstruction errors.

The results confirm that our hierarchical decomposition hypothesis is not merely a theoretical construct but provides practical advantages. By breaking down the complex 4×4 mosaicing into two manageable 2×2 sampling steps, our framework enables more stable and physically plausible reconstruction at each stage. This progressive approach effectively mitigates error propagation and allows the network to learn scale-consistent mappings that respect the underlying physics of the imaging process. The substantial performance improvement over the direct 4×4 baseline provides compelling evidence for the necessity and effectiveness of our hierarchical sampling decomposition strategy.

Table 5

Quantitative comparison between the proposed hierarchical 2×2 sampling decomposition and the direct 4×4 mosaicing approach on the real-world dataset. The symbols $\uparrow$ and $\downarrow$ indicate that higher and lower values correspond to better performance, respectively. The best result for each metric is highlighted in bold.

Configuration	QNR $\uparrow$	D_A $\downarrow$	D_S $\downarrow$
Direct	0.7856	0.1212	0.0374
Ours	0.9186	0.0313	0.0175

4.9. Computational Cost

The computational cost of different methods is reported in Table 6, where the number of parameters, FLOPs, and inference time are evaluated using mosaiced and PAN images with spatial sizes of 1020×1102 and 2040×2208 , respectively. As observed, EFN achieves the lowest computational cost due to its compact one-stage design, with only 0.10M parameters, 0.46T FLOPs, and 0.26s inference time. However, its reconstruction performance is inferior to the proposed method, especially on real-world data. Compared with EBVIF, our method has a comparable parameter scale and computational complexity, requiring 0.48M parameters and 1.42T FLOPs, with an inference time of 0.84s. Although slightly slower, the proposed method achieves clearly better reconstruction accuracy and robustness. Moreover, compared with the two-stage baselines, our method remains computationally efficient. For example, PMNet-based methods require substantially higher FLOPs, while WFANet-based combinations usually suffer from longer inference time. Overall, the proposed framework achieves a favorable balance between reconstruction performance and computational efficiency, demonstrating its practical potential for high-quality mosaiced and PAN image fusion.

5. Conclusion

In this paper, we present an imaging-informed progressive fusion framework for reconstructing HR HSIs from LR mosaiced and PAN observations. By explicitly modeling the hierarchical physics of mosaic degradation, our approach decomposes reconstruction into three physically grounded stages with cross-scale parameter sharing. The framework incorporates a three-stage composite loss that enforces intermediate and observation consistency to preserve structural relationships across scales. This physics-guided design transforms the highly ill-posed inverse problem into a series of tractable sub-tasks while maintaining interpretability. Extensive experiments demonstrate state-of-the-art performance without paired training data, achieving superior spectral fidelity and spatial detail while enhancing interpretability.

Table 6

Computational cost comparison of the competing fusion methods. Note that the tested mosaiced and PAN image here have a spatial dimension of 1020×1102 and 2040×2208 , respectively.

Method	Param (M)	FLOPs (T)	Time (s)
PPID*-VBPN	2.95	1.81	2.22
PPID*-PanGAN	0.19	0.89	2.27
PPID*-WFANet	0.53	1.82	7.04
LSAN-VBPN	3.24	2.08	0.46
LSAN-PanGAN	0.49	1.16	0.51
LSAN-WFANet	0.83	2.09	3.77
PMNet-VBPN	3.60	17.82	1.73
PMNet-PanGAN	0.85	16.90	1.44
PMNet-WFANet	1.18	17.83	5.45
EFN	0.10	0.46	0.26
EBVIF	0.17	1.41	0.77
Ours	0.48	1.42	0.84

* running on the CPU.

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Data Availability

The code and data used in the experiments are available at <https://github.com/Nan-Wong98/>.

CRedit authorship contribution statement

Nan Wang: Conceptualization, Methodology, Software, Writing - Original draft. **Fangshuo Xu:** Data curation, Validation, Investigation. **Renwei Dian:** Conceptualization, Methodology, Project administration, Funding acquisition. **Shutao Li:** Supervision, Review and editing.

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